

10 Audit-Readiness Gaps in Regulated IT and Automation Systems

What QA, CSV, and IT leaders should fix before a vendor audit, client audit, or regulatory inspection

About the Author

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Introduction

Most audit findings do not start with one dramatic failure.

They usually start with small, accepted weaknesses in validation logic, vendor oversight, documentation control, change management, data integrity, and system governance that become visible the moment an auditor asks for evidence.

I have seen these gaps across laboratory systems, PLC and SCADA environments, ERP and SAP platforms, BMS and EMS systems, data centres, and software vendors in regulated life-science settings across Europe, the US, India, APAC, and MENA.

This guide highlights 10 recurring weaknesses that make systems look operational on paper but difficult to defend during an audit or inspection.

For each gap, I focus on three practical questions:

- **What auditors see** — The visible weakness
- **Why it matters** — The compliance risk
- **What good looks like** — The evidence of control

Use this as a quick internal review tool before a regulatory inspection, supplier qualification exercise, client audit, or remediation program.



Figure 1: Figure 1: Best Practices for Conducting a Compliance Audit

Gap 1: Undefined Validation Decisions

What auditors see:

Teams cannot clearly explain why a system is classified as GxP or non-GxP, even though that decision drives the whole validation approach.

Why it matters:

If the rationale is weak, the control strategy usually becomes weak as well, and auditors quickly question whether the system was validated to the right level or at all.

What good looks like:

A documented risk-based decision supported by intended use, impact assessment, system complexity, and traceable quality rationale. The decision should justify the validation scope and ongoing control level.

Gap 2: Weak Vendor Assessment Records

What auditors see:

Many organizations rely heavily on vendors, but the impact, complexity, and GxP relevance of those vendors are not documented well enough to justify oversight decisions.

Why it matters:

Vendor assessments and remote IT GxP vendor audits are now central to modern compliance, especially for platforms like Veeva, ServiceNow, Benchling, ValGenesis, SharePoint, Google Apigee, Siemens Automation, Rockwell Automation, and other regulated technology providers. Weak vendor records suggest poor supply-chain control.

What good looks like:

A current vendor assessment package that explains risk category, service scope, audit history, quality controls, data handling, change management, and justified oversight level based on GxP impact.

Gap 3: Incomplete SDLC and Configuration Evidence

What auditors see:

Teams often have documents, but not a connected story between CMDB records, SDLC deliverables, validation evidence, and

operational ownership.

Why it matters:

Auditors do not just ask whether documents exist; they ask whether the records align and support a controlled lifecycle from planning through retirement.

What good looks like:

Configuration records, system inventory, validation documents, approvals, change history, and support ownership all match and can be traced quickly. Evidence is organized so QA, IT, engineering, and auditors see the same controlled story.

Gap 4: Overdue Periodic Reviews

What auditors see:

Periodic reviews are frequently planned but not executed on time, or they are completed as a checkbox exercise with little real evaluation or evidence of review findings.

Why it matters:

An overdue or weak review suggests the organization has lost active control of the system after go-live. Regulators expect continued oversight proportional to risk.

What good looks like:

A periodic review that checks system state, current access, open deviations, recent changes, vendor updates, backup and recovery posture, security patches, and document currency. The review should be timely, documented, and result in actionable findings.

VALIDATION LIFECYCLE

PLANNING



SPECIFICATION



VERIFICATION



REPORTING

Figure 2: Figure 2: System Validation Lifecycle - Planning, Specification, Verification, and Reporting

Gap 5: Poor Change Control Discipline

What auditors see:

Changes are made to software, infrastructure, interfaces, or configuration without enough impact assessment, review, approval, or traceable evidence of controlled implementation.

Why it matters:

Uncontrolled change is one of the fastest ways to damage a previously valid state. Auditors expect a clear CR trail showing what changed, why, who approved, what testing was done, and proof the change was implemented.

What good looks like:

Every change request includes impact analysis, testing expectations, approval route, implementation evidence, post-change verification, and formal closure. Changes are tracked, approved before implementation, and verified after.

Gap 6: Weak Migration, DR, and Retirement Controls

What auditors see:

Data migration plans, disaster recovery reports, or system retirement records are often incomplete, out of date, or missing entirely, especially for older systems and infrastructure.

Why it matters:

Gaps in migration, DR, or retirement raise immediate questions about data protection, continuity planning, and record integrity at critical lifecycle stages.

What good looks like:

Approved migration plans and reports showing data validation, testing results, and successful transfer. Tested DR evidence showing

recovery is feasible. Formal retirement documentation showing the system was controlled to end-of-life and records were preserved or destroyed appropriately.

Gap 7: Deviation and CAPA Closure Gaps

What auditors see:

Operational issues are sometimes logged, but investigations are shallow, CAPAs are weak, or closure evidence does not prove that the root cause was actually addressed or that recurrence was prevented.

Why it matters:

In regulated environments, unresolved or weakly investigated failures suggest that quality events are being managed administratively rather than scientifically. Auditors look for real corrective actions and evidence of effectiveness.

What good looks like:

Deviations, failure investigations, and CAPAs are timely, risk-based, and supported by evidence that the action actually prevented recurrence. Closure includes verification that the fix works and will be sustained.

Gap 8: CSV Documentation That Exists But Is Hard to Defend

What auditors see:

Many teams have URS, FS, DS, SRS, VMP, IQ, OQ, PQ, FRA, RAP, and summary reports, but the package is inconsistent, outdated, disconnected from actual system use, or difficult to navigate during an audit.

Why it matters:

During an audit, weak documentation makes even a technically sound system look poorly governed. Auditors need to see clear

requirements, traceability, testing evidence, and sign-off.

What good looks like:

Documentation is current, internally consistent, approved, risk-based, and clearly tied to intended use, system design, testing, release criteria, and ongoing control. All documents can be located quickly and cross-referenced.

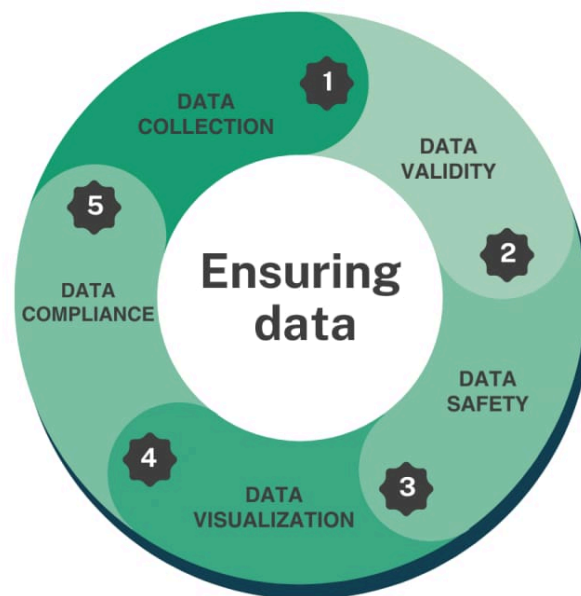


Figure 3: Figure 3: Data Management Cycle - Collection, Validity, Safety, Visualization, and Compliance

Gap 9: Data Integrity Blind Spots

What auditors see:

Organizations often say data integrity is important, but their controls do not fully show ALCOA+ thinking, audit trail discipline, or reliable traceability under pressure. Evidence quality is inconsistent.

Why it matters:

The core question is: *"Is the data real—and will a regulator trust it?"* Auditors judge readiness by examining records, audit trails, and

operational evidence. Weak controls here create major risk even if systems are technically sound.

What good looks like:

Data is attributable, legible, contemporaneous, original, accurate, and supported by access control, user training, review discipline, and defensible evidence practices. Audit trails capture who, what, when, and why. Data backups and recovery are tested.

Gap 10: Automation Systems Treated Like Black Boxes

What auditors see:

PLC, SCADA, BMS, EMS, MCC, laboratory equipment, access control, fire alarm systems, ERP, SAP, serialization, and infrastructure are often running successfully, but the compliance evidence around them is fragmented, incomplete, or weak.

Why it matters:

These platforms are deeply embedded in regulated operations. Weak governance around them creates major audit exposure even when production continues normally. Auditors expect the same level of control as software systems.

What good looks like:

Automation and IT systems are documented, assessed, validated, reviewed, and maintained in a way that lets QA, IT, engineering, and auditors all see the same controlled story. Change history, periodic review, and incident response are traceable.

Key Takeaways

Most companies do not need more documents. They need:

- Clearer decisions about GxP scope and control levels
- Better evidence of that scope and the controls that support it

- Stronger lifecycle control from planning through retirement
- A more realistic view of where audit risk actually sits
- Practical tools to verify readiness before the auditor arrives

Internal Readiness Checklist

Use this guide as your internal checklist. For each gap, ask:

- ✓ Is this weakness present in our environment?
- ✓ How would an auditor see it?
- ✓ What evidence would close it?
- ✓ Do we have a timeline to address it?

Next Steps

Need an independent review of your regulated IT, automation, vendor, or CSV controls?

Contact Kushal Vyas for:

- Audit-readiness assessments
- Vendor audit support
- CSV and IT quality reviews
- Regulatory inspection preparation

Contact Information

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Document Information

About This Guide

This guide is based on 13+ years of GxP audit, CSV, IT quality, and regulatory inspection experience across manufacturing sites, IT vendors, data centres, and life-science organizations in Europe, the US, India, APAC, and MENA.

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References

This guide synthesizes industry best practices from:

- FDA 21 CFR Part 11 and ICH guidelines
- GAMP 5 framework for IT system validation
- PDA Technical Report 60 on computer system validation
- ISPE Baseline guides for pharmaceutical systems
- ISO 14644 cleanroom standards
- Modern DevOps and IT compliance frameworks for life-science operations